

# EXPERIMENT 15

## GIT BRANCHING AND MERGE CONFLICT RESOLUTION

**Aim:** To demonstrate how to work with Git branches and resolve merge conflicts when collaborating with others.

### Procedure

1. Clone the shared Git repository to the local machine.
2. Create a new feature branch and switch to the branch.
3. Make changes to a file such as README.md or a specific function in the source code.
4. Commit the changes made on the feature branch.
5. Push the feature branch changes to the remote repository.
6. Pull the latest changes from the main branch before merging.
7. Switch back to the feature branch.
8. Merge the main branch into the feature branch.
9. If a merge conflict occurs, manually edit the conflicting file and choose the correct content.
10. Mark the resolved file as resolved and commit the merge.
11. Push the updated feature branch to GitHub and create a pull request.

### Conflict Encountered

A merge conflict occurred in the README.md file in the **EX15 Collaboration** section. The conflicting versions contained different descriptions of how the change was made.

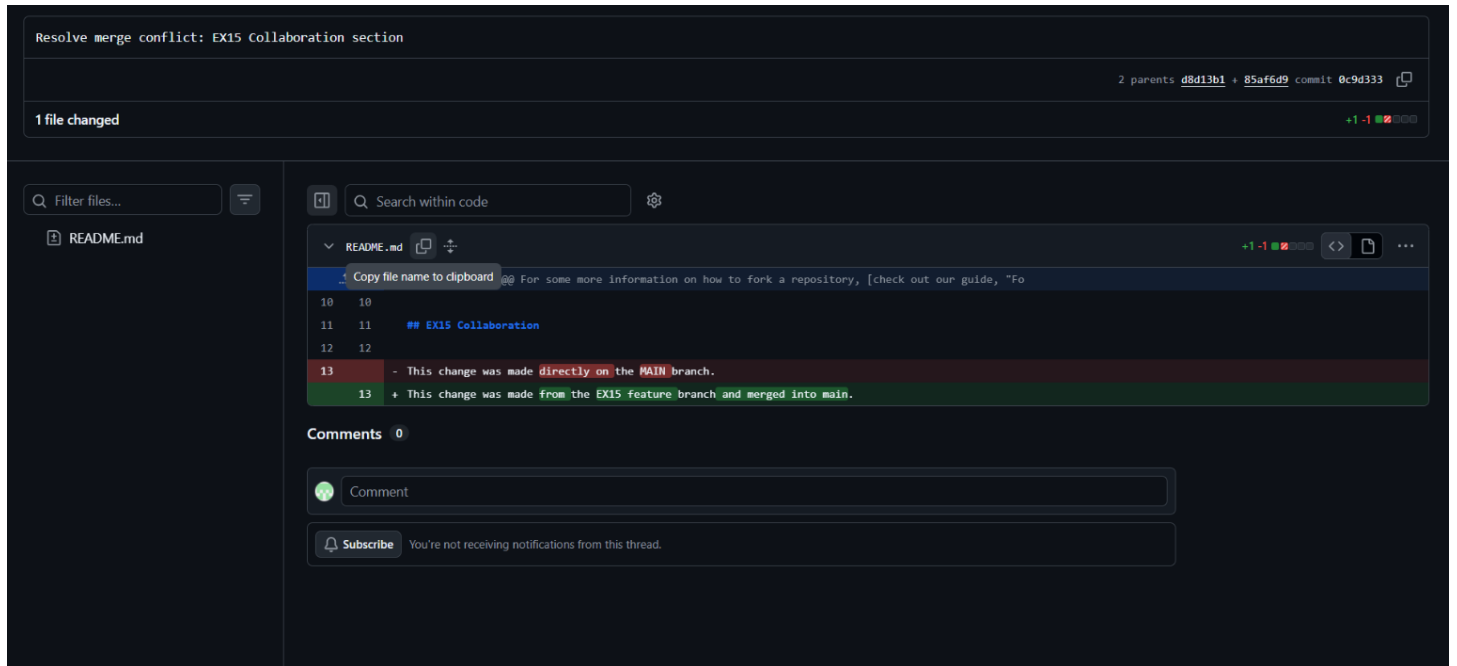
### Conflict Resolution

1. The conflicting README.md section was opened and the conflict markers were reviewed.
2. The incorrect/direct-main-branch wording was removed.
3. The correct feature-branch wording was retained: the change was made from the EX15 feature branch and merged into main.
4. The conflict was marked as resolved.
5. The resolved merge was committed and the updated branch was prepared for pushing to GitHub.

### Summary

Git branching was used to isolate changes during collaborative development. The latest main branch was merged into the feature branch before the pull request. When the README.md conflict occurred, the conflicting content was manually reviewed and corrected. The resolved file was committed and the feature branch was pushed to GitHub for collaboration and pull-request review.

# SCREENSHOT OF COMPLETED MERGE CONFLICT RESOLUTION



**Result:** The Git branch merge and conflict-resolution workflow was successfully demonstrated, and the conflicting `README.md` changes were resolved.